NC STATE UNIVERSITY

# ECE 592-091 - Resonant Power Converters

# Lab 3 - Analysis and Simulation of a Class D Series-Resonant Converter

Sanjana Ashok Bhalekar
Raghav Chandrasekaran
Sydney Cheng
Abhishek Ajit Deshpande

Professor - Dr. Zeljko Pantic
Teaching Assistance - Zhansen Akhmetov

# Table of Contents


| | |
|---|---|
| Objective | 3 |
| Task 1 | 5 |
| Task 2 | 8 |
| Task 3 | 9 |
| Task 4 | 11 |
| Task 5 | 15 |
| Task 6 | 24 |
| Task 7 | 26 |
| Appendix A | 28 |
| Appendix B | 31 |

# Objective

Objective The objective of this lab is to allow students to conduct an in-depth analysis of the Series-Resonant Converter (SRC), the same one that will be tested and controlled in Labs 4-6. The simulations will allow some operation conditions and curves to be calculated that are impossible in the experimental setup with the existing equipment.

The electric scheme of the proposed Class D SRC is presented in Fig. 1. The resonant converter consists of a DC voltage source, a half-bridge switching network, a series-resonant network, a center-tapped transformer, a diode rectifier, a capacitive filter, and a resistive load. The resonant capacitor consists of MLCC C0G (NP0) capacitors connected in parallel. The filter capacitance is achieved by combining larger aluminum electrolytic capacitors and smaller ceramic capacitors with high self-resonant frequency. The converter's input voltage ranges from 36 V to 60 V, with VIN,n=44 V as the nominal value. The converter maintains a nominal VO,n=12 V, at the output for the nominal load resistance of RL,n=2.4 Ω. The resistive load can vary from 1.92 Ω to 9.6 Ω, causing the output power to vary from 15 W (25% of the nominal power) to 75 W (125% of the nominal power). The input and output specifications are summarized in Table 1. Table 2 contains the specifications of other circuit components, including switches, resonant and filter capacitors, rectifier diodes, the transformer, and the inductor. The converter will utilize frequency control to regulate the output voltage at 12 V against the load and input voltage variation.

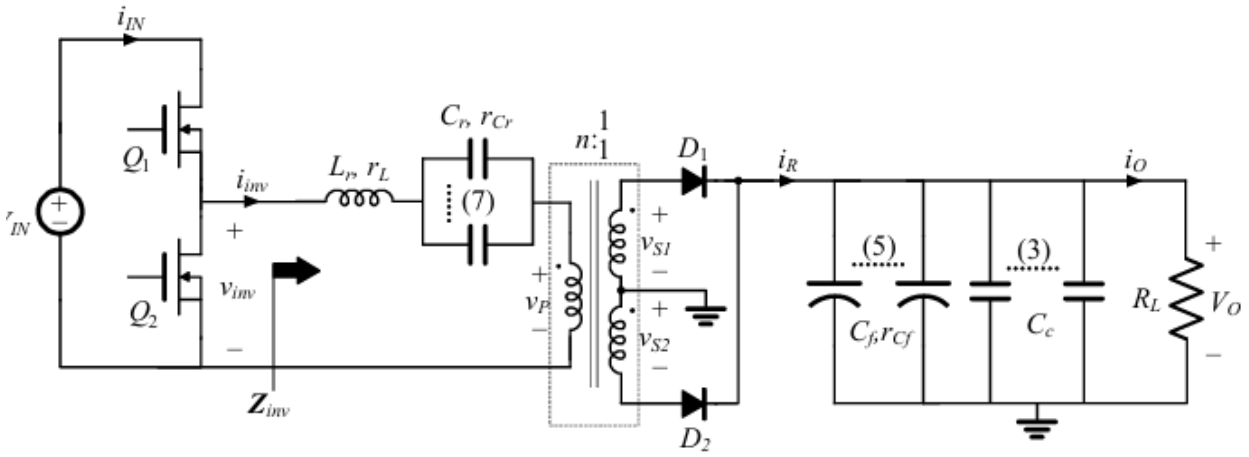



Fig. 1: Circuit diagram of the Class D SRC.

Table 1: Converter Input and Output Specifications

| Parameter | Minimum | Maximum | Nominal |
|---|---|---|---|
| Input voltage $V_{IN}$ [V] | 36 | 60 | 44 |
| Output Voltage $V_O$ [V] | 12 | 12 | 12 |
| Resistive Load $R_L$ [Ω] | 1.92 | 9.6 | 2.4 |
| Output Power $P_O$ [W] | 15 | 75 | 60 |
| Forecasted rectifier efficiency $\eta_R$ [%] | - | - | 97 |
| Forecasted transformer efficiency $\eta_{tr}$ [%] | - | - | 100 |
| Forecasted inverter efficiency $\eta_I$ [%] | - | - | 97 |
| Maximum Ripple $V_{o,ripple}$ [%] | - | 1% | - |
| Auxiliary Input Voltage [V] | - | - | 12 |
| Intended resonant frequency $f_0$ [kHz] | - | - | 100 |
| Gate resistance $R_g$ [Ω] | - | - | 10 |

| Component | Value |
|---|---|
| MOSFET Switches | 100 V, 90 A |
| Rectifier Diodes | 45 V, 30 A |
| Resonant Capacitance $C_r$ | 4 x 5.6 nF<br>\|\|<br>3 x 6.8nF<br>Total:42.8 nF |
| Resonant Inductance $L_r$ | 59.2 uH |
| Transformer ratio $n$ | 1.2 |
| Filter Capacitors $C_f$ and $C_c$ | 0.1 nF<br>10 nF<br>1 uF<br>5 x 33uF<br>165 uF |
| Gate Driver Voltage $V_{dr}$ | 0 - 5.5 V |
| Half-Bridge Deadtime | 155.7 ns |

# Task 1

Considering the forecasted efficiencies of the inverter and rectifier in Table 1, derive the expression of the converter's voltage transfer function (DC-DC voltage gain) MV as the function of frequency f. Use the derive expression to plot the gain as a function of frequency f for three different loads RL,min, RL,n, and RL,max and the frequency range from 60% to 140% of the resonant frequency.

The efficiency of the half-bridge SRC with a transformer center-tapped rectifier, shown in Fig. 15.2(b), is obtained from (2.72), (2.73), and (6.69) as

$$\eta = \eta_{Ir}\eta_R = \frac{\eta_{tr}}{1 + \frac{V_F}{V_O} + \frac{\pi^2 R_F}{8R_L} + \frac{\pi^2 r \eta_{tr}}{8n^2 R_L} + \frac{r_C}{R_L}\left(\frac{\pi^2}{8} - 1\right)}. \qquad (15.5)$$

$$M_V = \frac{V_O}{V_I} = M_{VI}|M_{VR}| = \frac{\eta}{2n\sqrt{1 + Q_L^2\left(\frac{\omega}{\omega_o} - \frac{\omega_o}{\omega}\right)^2}}$$

$$Q_L = \frac{\omega_o L}{R_i + r}$$

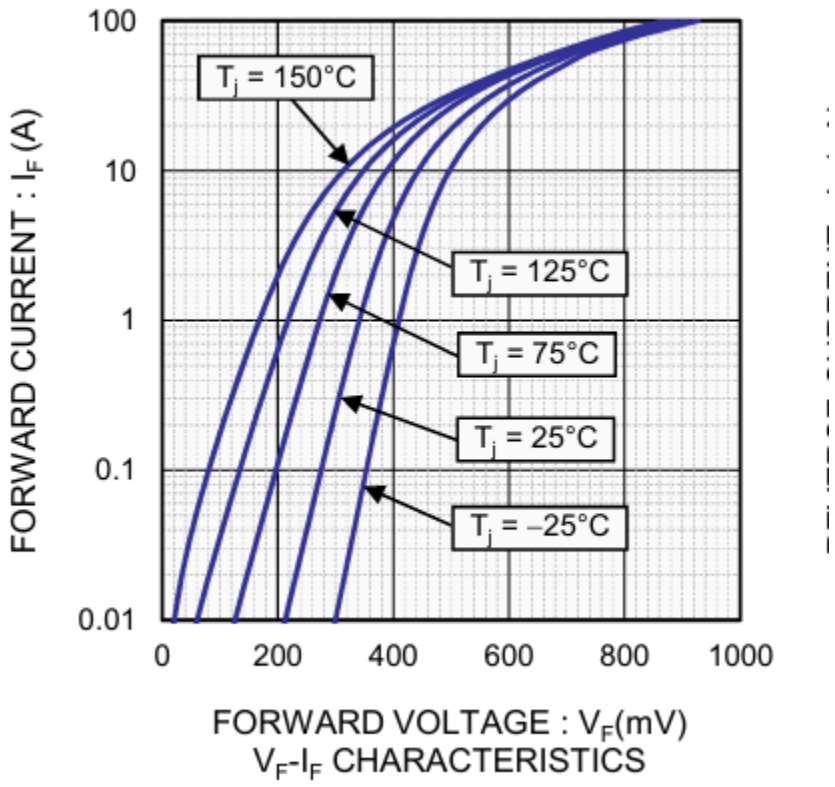
$$r_{half-bridge} = r_{DS} + r_L + r_{Cr}$$

$$r_{full-bridge} = 2r_{DS} + r_L + r_{Cr}$$

Since this is a center-tapped configuration, we use the second formula for the equivalent r value.

$$R_{i,FW(CT)} = \frac{8n^2 R_L}{\pi^2 \eta_R}$$

Using these formulas, we can plot the graph in Matlab by sweeping the frequency from **40% to 160%** at three different r loads( RL,min, RL,n, and RL,max).




Rf of the diode was found using the slope at 25 degrees Celsius

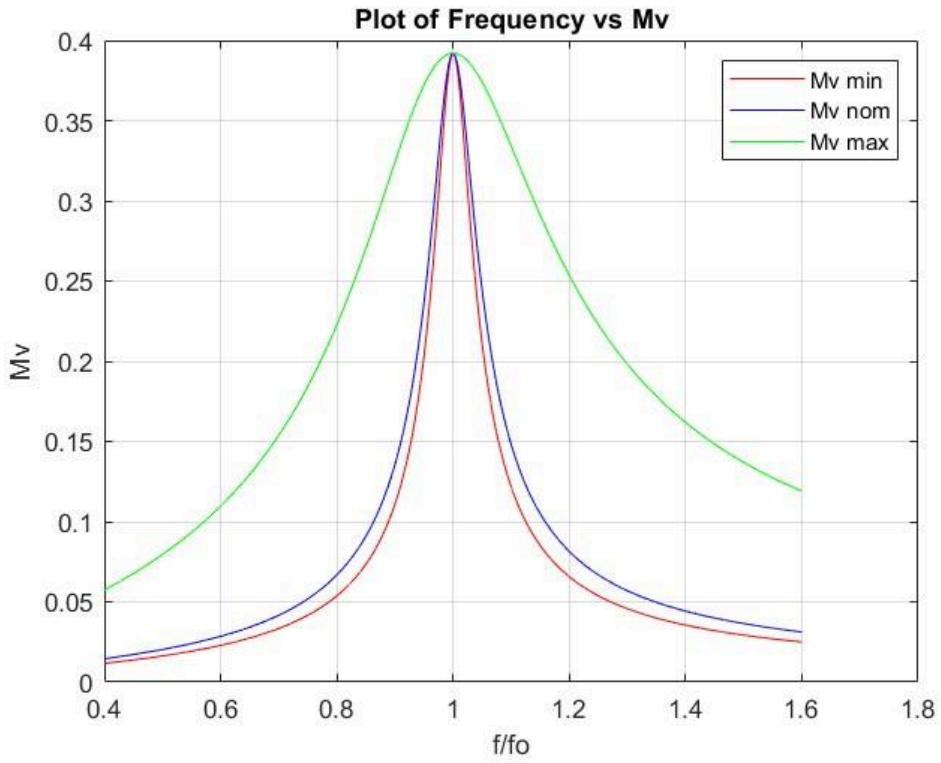
Rf = 0.0175 ohms.

From the MATLAB code, the Ri values were found to be:

$R_{i,min}$=2.3104Ω

$R_{i,n}$=2.8880Ω

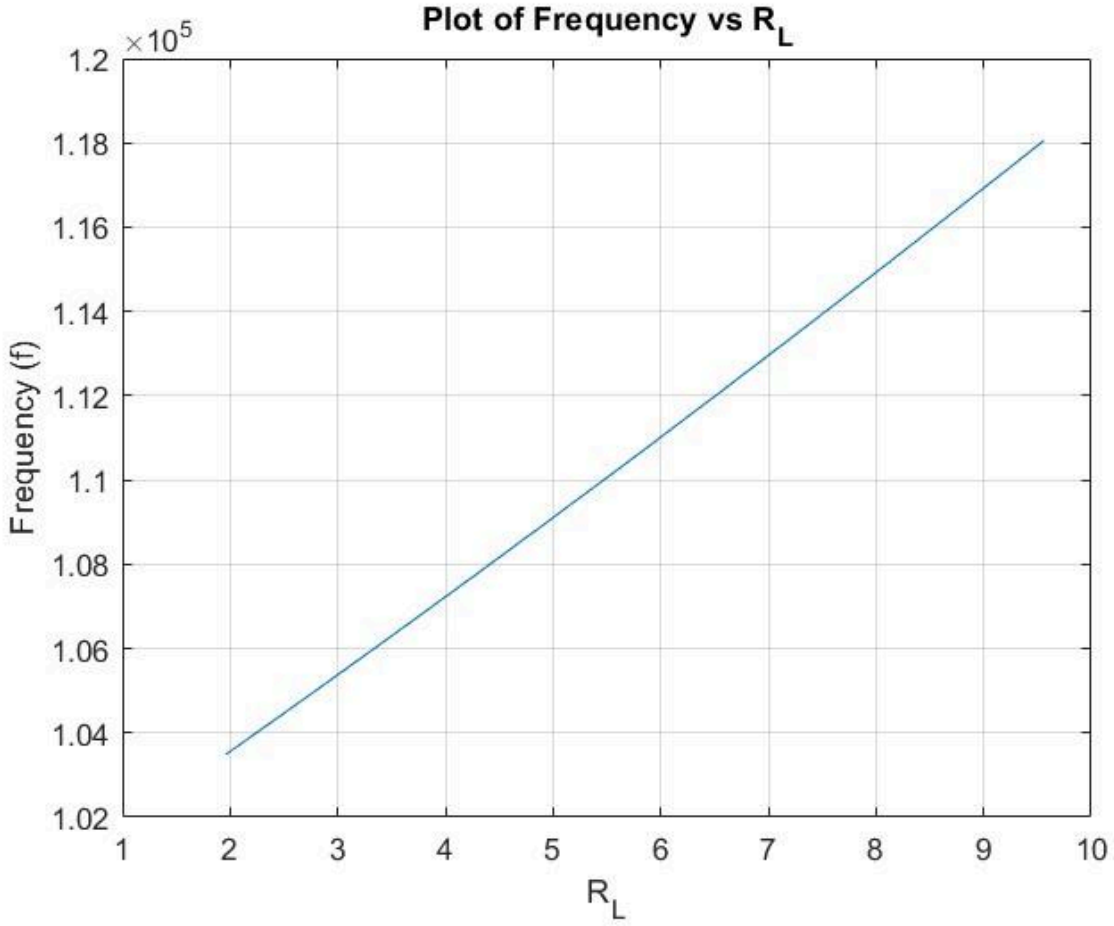
$R_{i,max}$=11.5519Ω




The graph shows that the voltage gain peaks at the resonant frequency and decreases symmetrically when the frequency deviates from this point; different load values affect the sharpness of the curve

# Task 2

Considering the nominal values of input and output voltages (VIN,n=44 V, VO,n=12 V), derive, calculate, and plot the frequency variation as a function of load resistance for the range of load resistances from RL,min=1.92 Ω to RL,max=9.6 Ω. What frequency range should the converter cover to support the full load range variation? Include the derivation, the plot, and the discussion in your report.

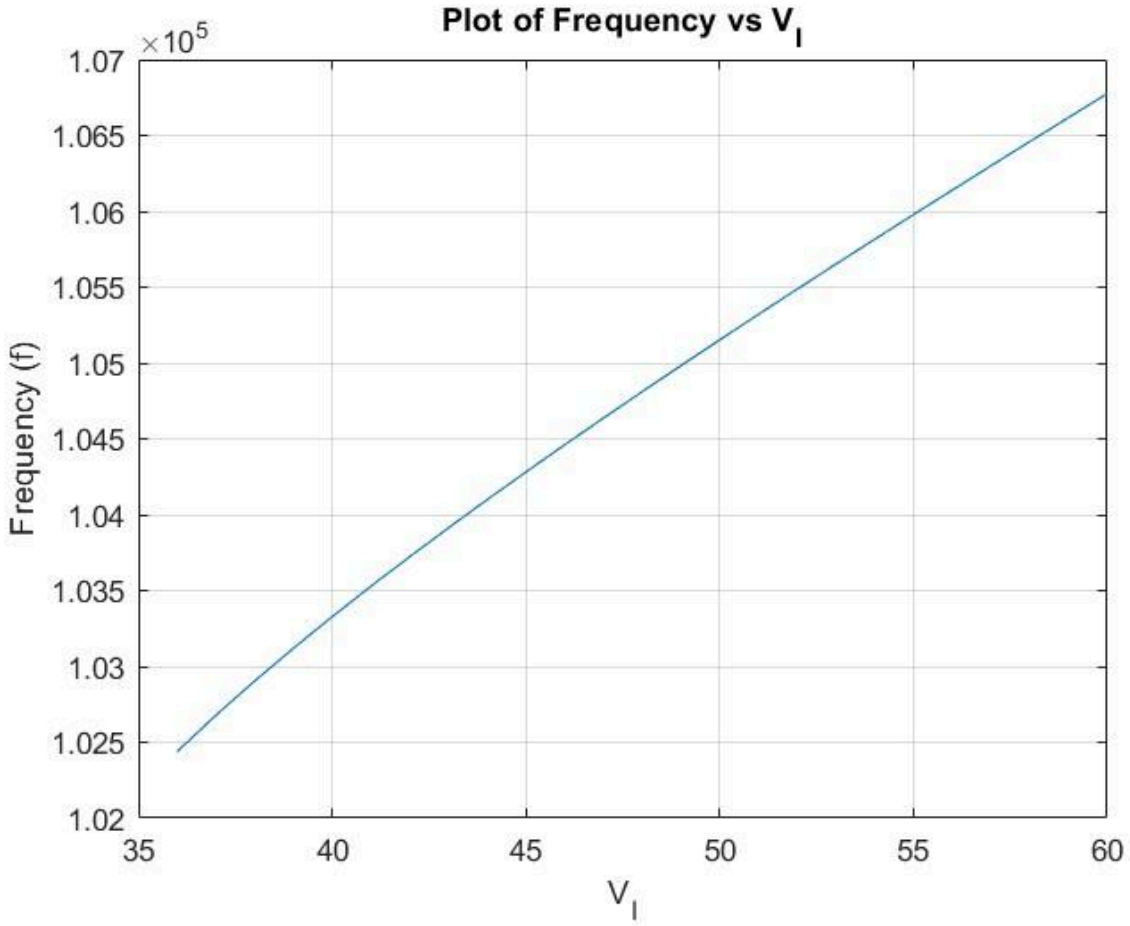



The simulation indicates that as the load resistance increases, the operating frequency must be adjusted upward to maintain the nominal output voltage, highlighting a non-linear relationship.

The frequency for Rl = 2.4 ohms is **104300 Hz.**

# Task 3

Considering the nominal values of the output voltage and the output load ($V_{O,n}$=12 V, $R_{L,n}$=2.4 Ω), derive, calculate, and plot the frequency variation as a function of the input voltage VIN varying in the range from $V_{IN,min}$=36 V to $V_{IN,max}$=60 V. What frequency range should the converter cover to support the full input voltage range variation? Include the derivation, the plot, and the discussion in your report.




The plot reveals that increasing the input voltage requires a corresponding increase in operating frequency to keep the output voltage constant due to the converter's inherent voltage-frequency characteristics.

From task 2 and task 3 plots the Values of the frequency are written below in the table. For this analysis, the ESR of the Inductor and the Capacitor were considered 0, but the R_ds of the MOSFET was considered

| **R_L values(Ω)** | **Frequency(Hz)** | **V_I (V)** | **Frequency (Hz)** |
|---|---|---|---|
| 1.92 | 103413 | 36 | 102438 |
| 2.4 | 104300 | 40 | 103327 |
| 2.92 | 105230 | 44 | 104099 |
| 3.42 | 106150 | 48 | 104811 |
| 4.42 | 108012 | 52 | 105487 |
| 6.92 | 112799 | 56 | 106139 |
| 9.6 | 118136 | 60 | 106766 |

# Task 4

In this task, the teams should build an LTSpice model of the studied converter. To do so, they should utilize the models provided by manufacturers and, when unavailable, create the models using the datasheet information. The losses in the transformer can be modeled by adding 20 m resistors into the primary and both secondaries. For modeling the inductor, adopt that its ESR is 40 mΩ. Use the following link to the manufacturer's website to locate the resonant capacitor from Table 2, and adopting a constant ESR value, determine its ESR from the curve provided there. Assign the nominal input voltage and nominal load and adjust the frequency in the open-loop mode so that the output voltage has the nominal value, too. Plot the five signals marked in Fig. 1: vinv (t), iinv(t), vp(t), iR(t), and vO(t) over three signal periods. As part of your report, submit the model and the five mentioned plots. The Figure below shows the overall LTSpice model the team used.

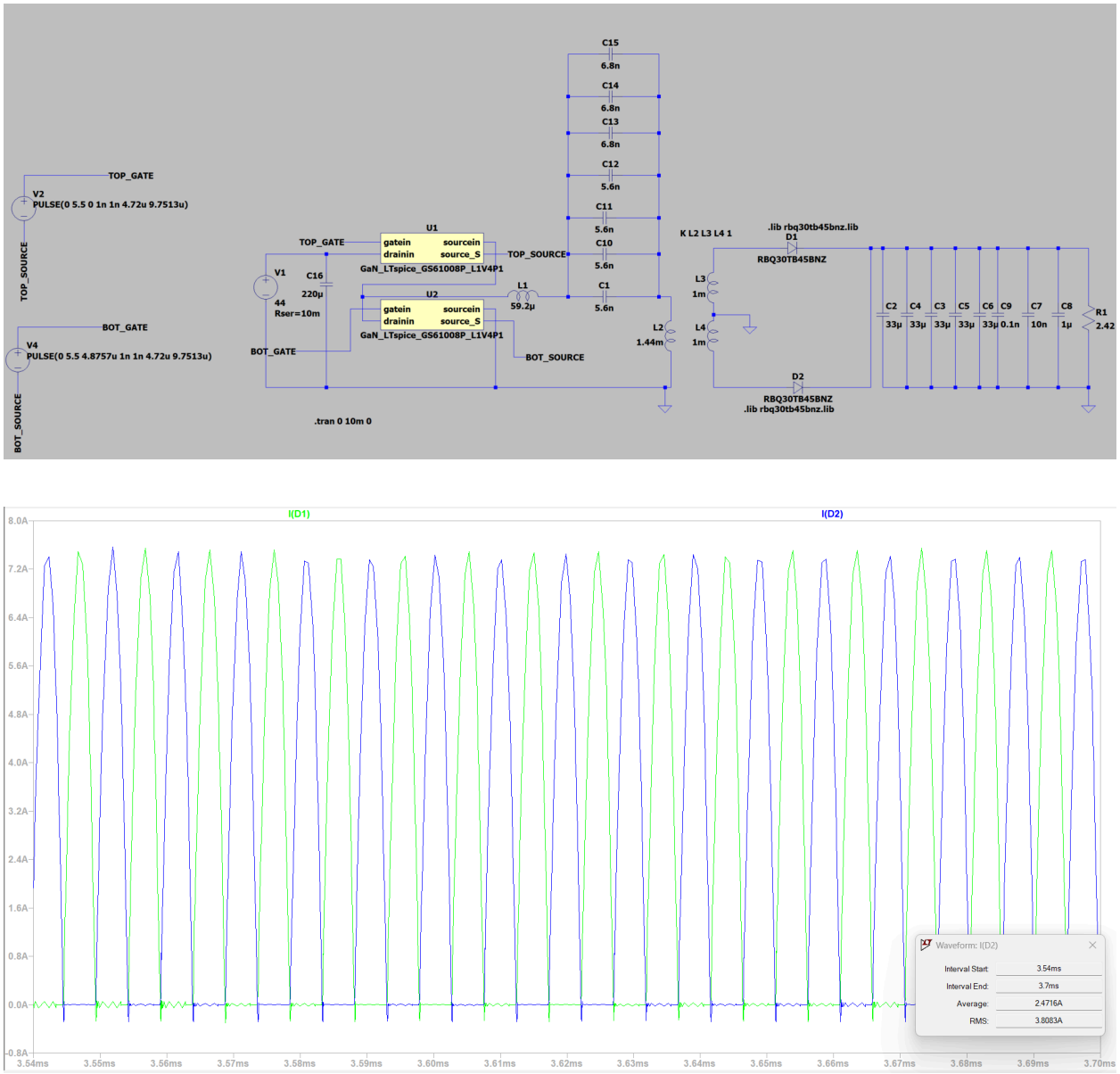






*Figure 4.1: I_r(t)*

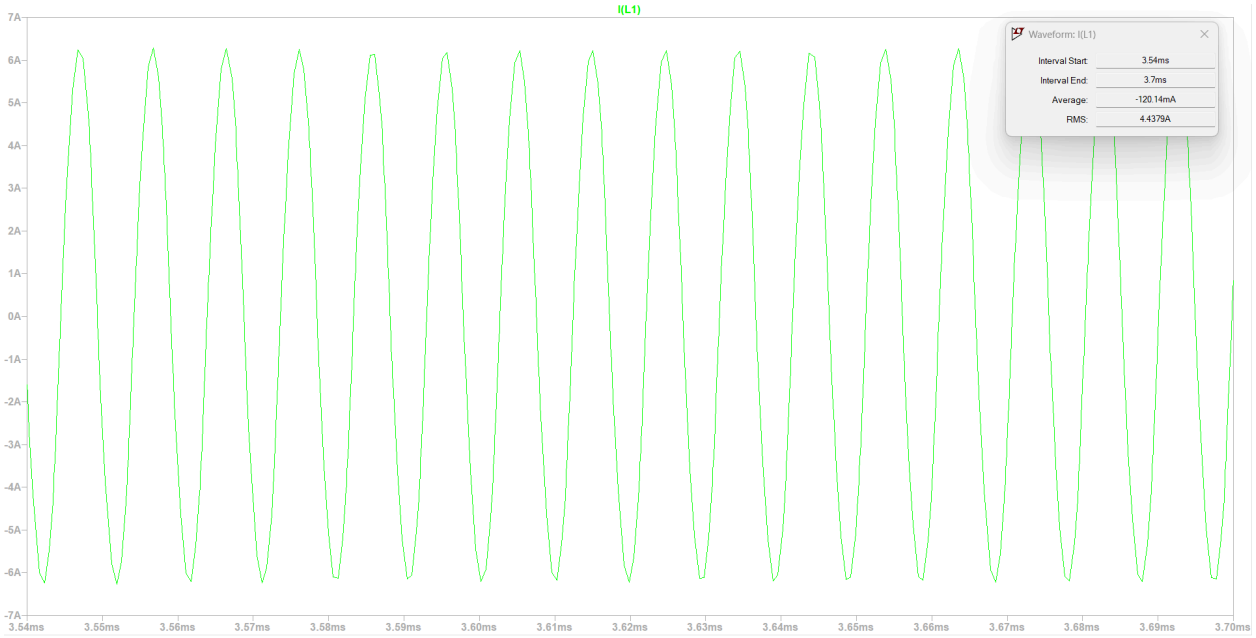



*Figure 4.21: I_inv(t)*




*Figure 4.3: V_inv(t)*

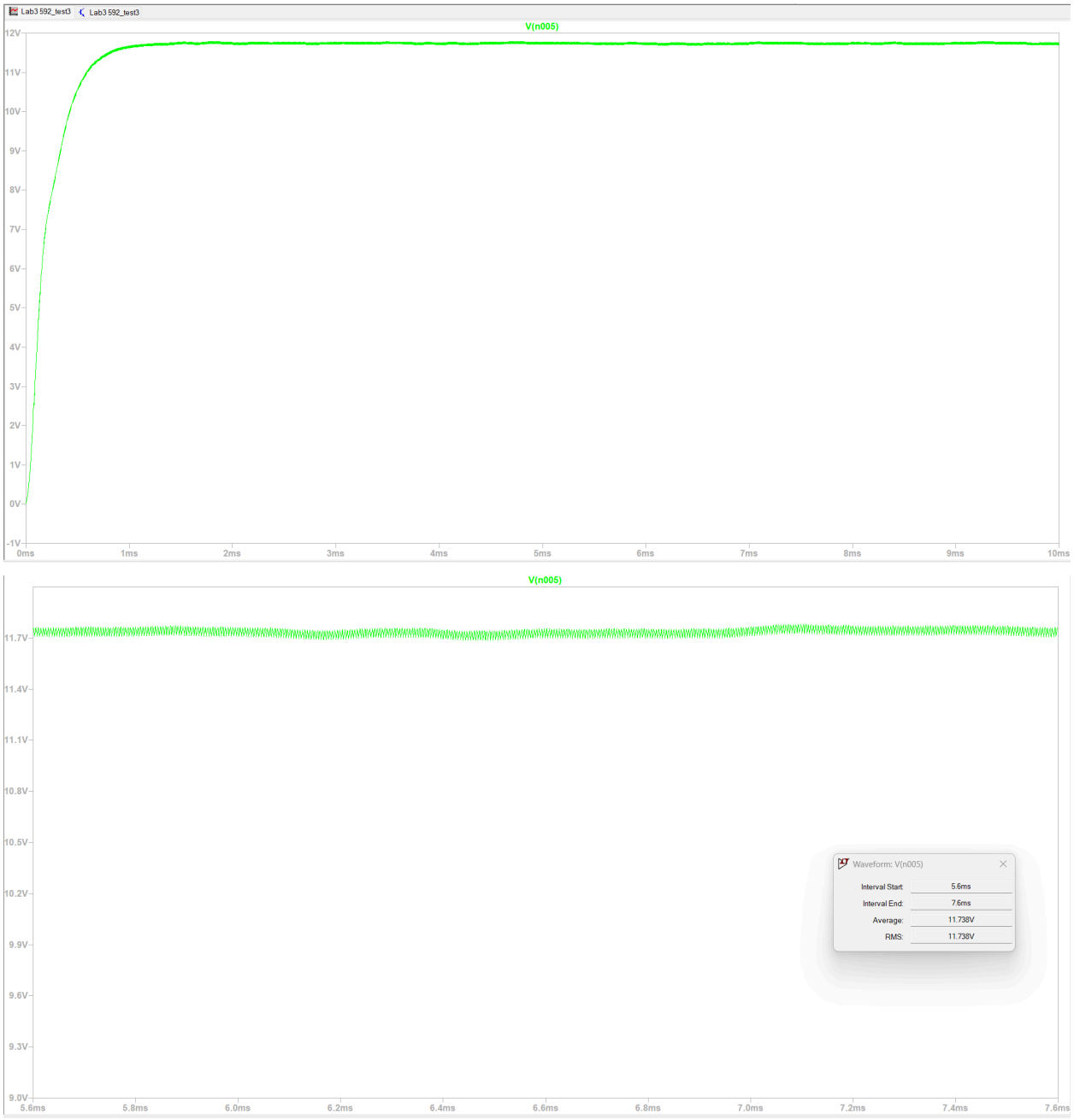


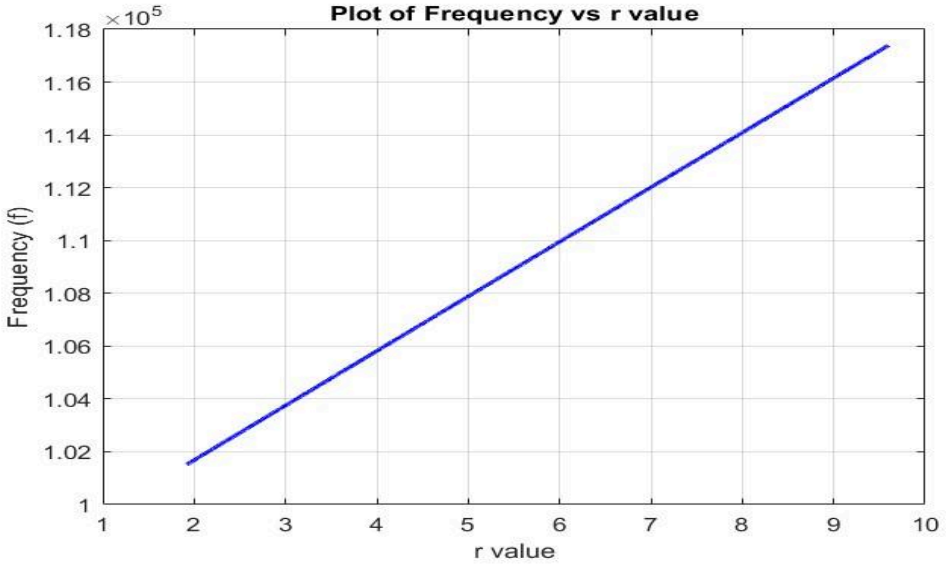

*Figure 4.4 &4.5: V_o(t) & V_o(t) zoomed*




*Figure 4.5: V_p(t)*

# Task 5

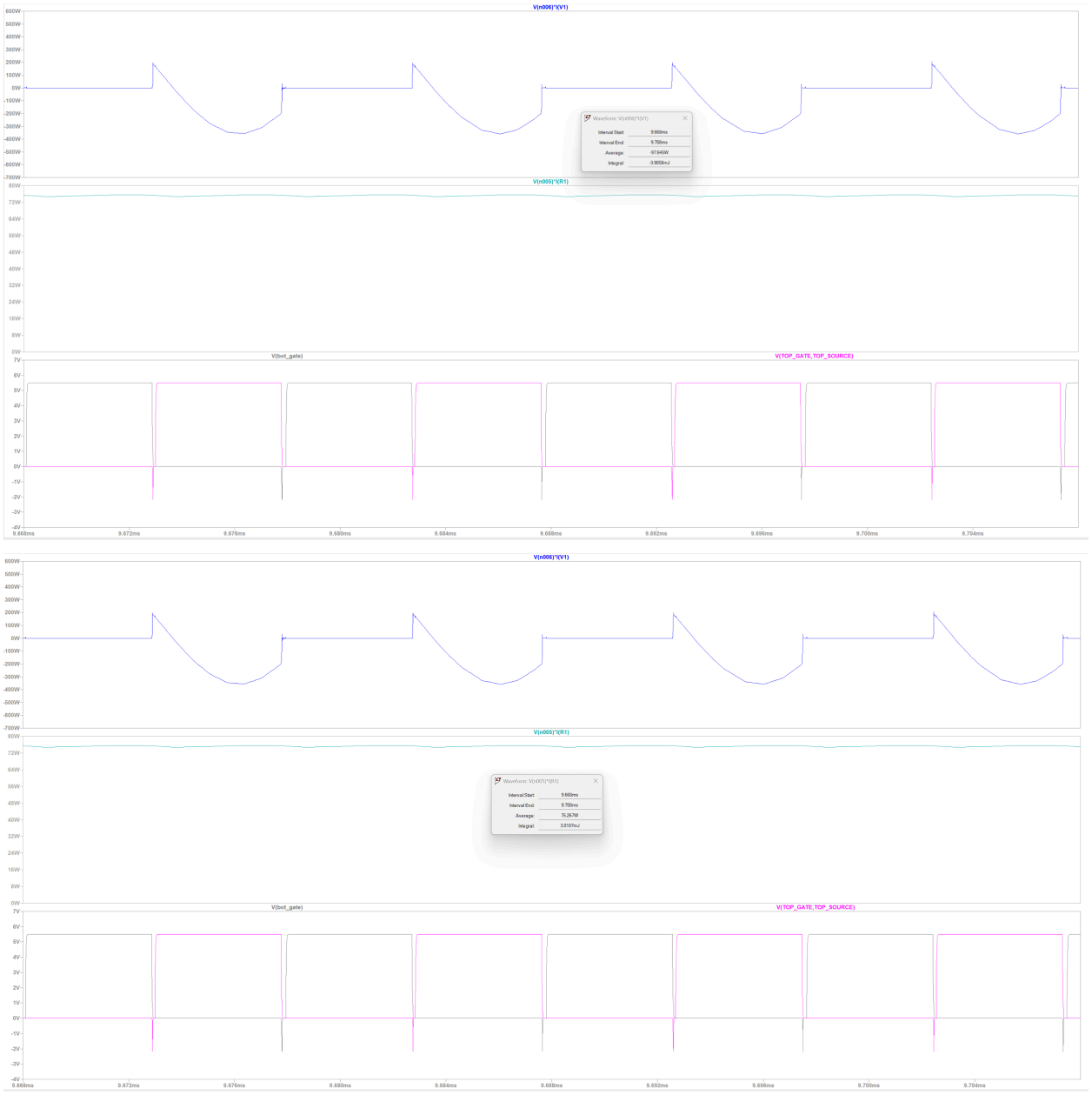
Use the model from Task 4 to repeat the analysis from Task 2. For VIN=VIN,n and selected load resistances RL, the frequency f should be varied in the open-loop mode until the nominal output voltage is reached VO=VO,n. To achieve uniform testing, utilize the following load values: RL=[1.92 2.42 2.92 3.42 4.42 6.92 9.6] Ω.



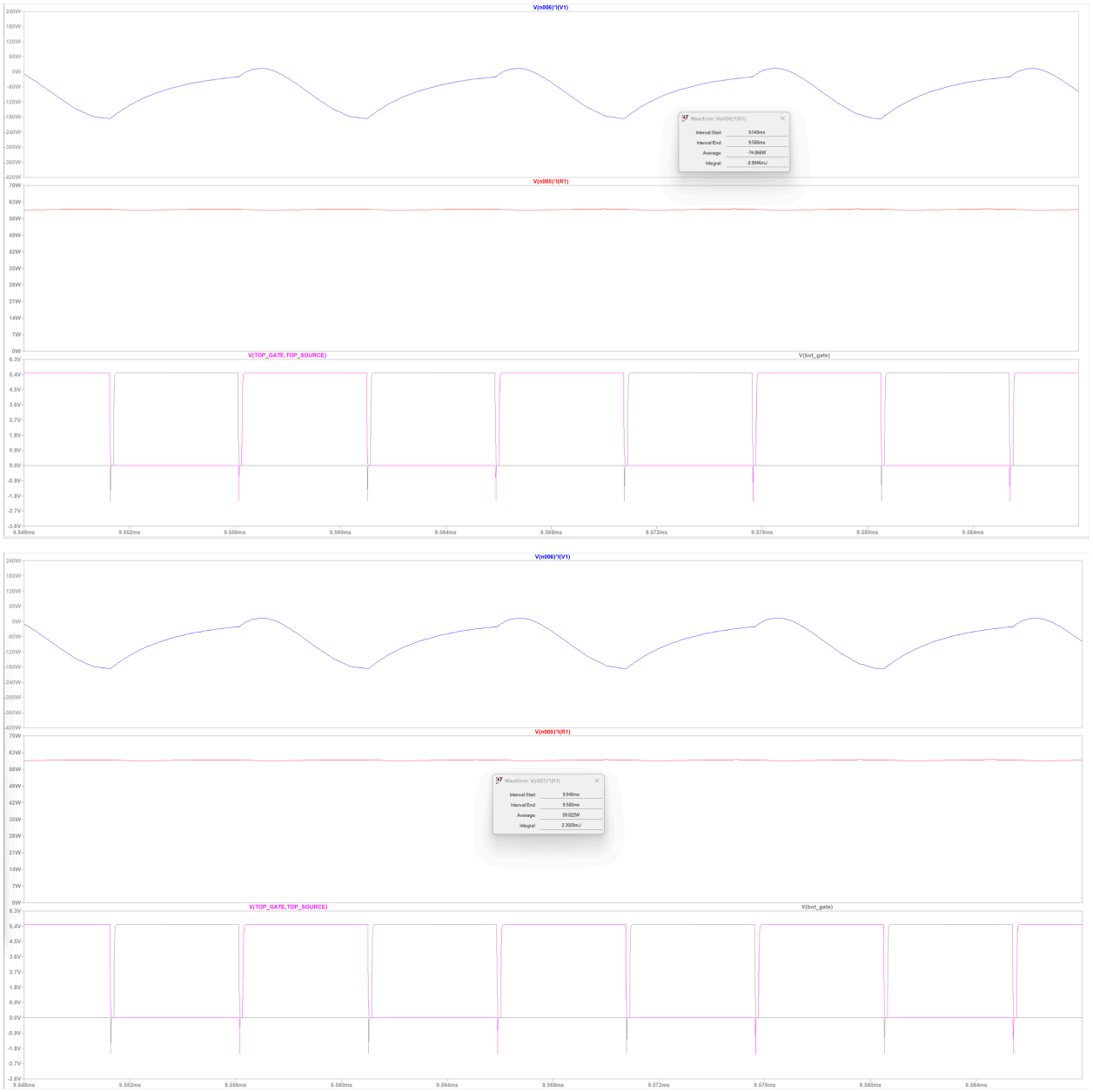

The plot of switching frequency f versus load resistance RL is shown above. The data represents the variation in switching frequency required to maintain nominal output voltage as the resistive load changes from RLmin=1.92 Ω to RLmax=9.6 Ω. The switching frequency increases as the load resistance increases, indicating that the converter adjusts its operating frequency to sustain the desired output voltage.

| Sr.no | Resistance (load) | Frequency |
|---|---|---|
| 1. | 1.92 | 101520 |
| 2. | 2.42 | 102550 |
| 3. | 2.92 | 103585 |
| 4. | 3.42 | 104615 |
| 5. | 4.42 | 106685 |
| 6. | 6.92 | 111854 |
| 7. | 9.6 | 117392 |

Table: Frequency values for varying load Resistance to maintain the output voltage as 12 V




*Figures 5.1 & 5.2: Input and Output powers at 1.92Ω*




*Figures 5.3 & 5.4: Input and Output powers at 2.42Ω*



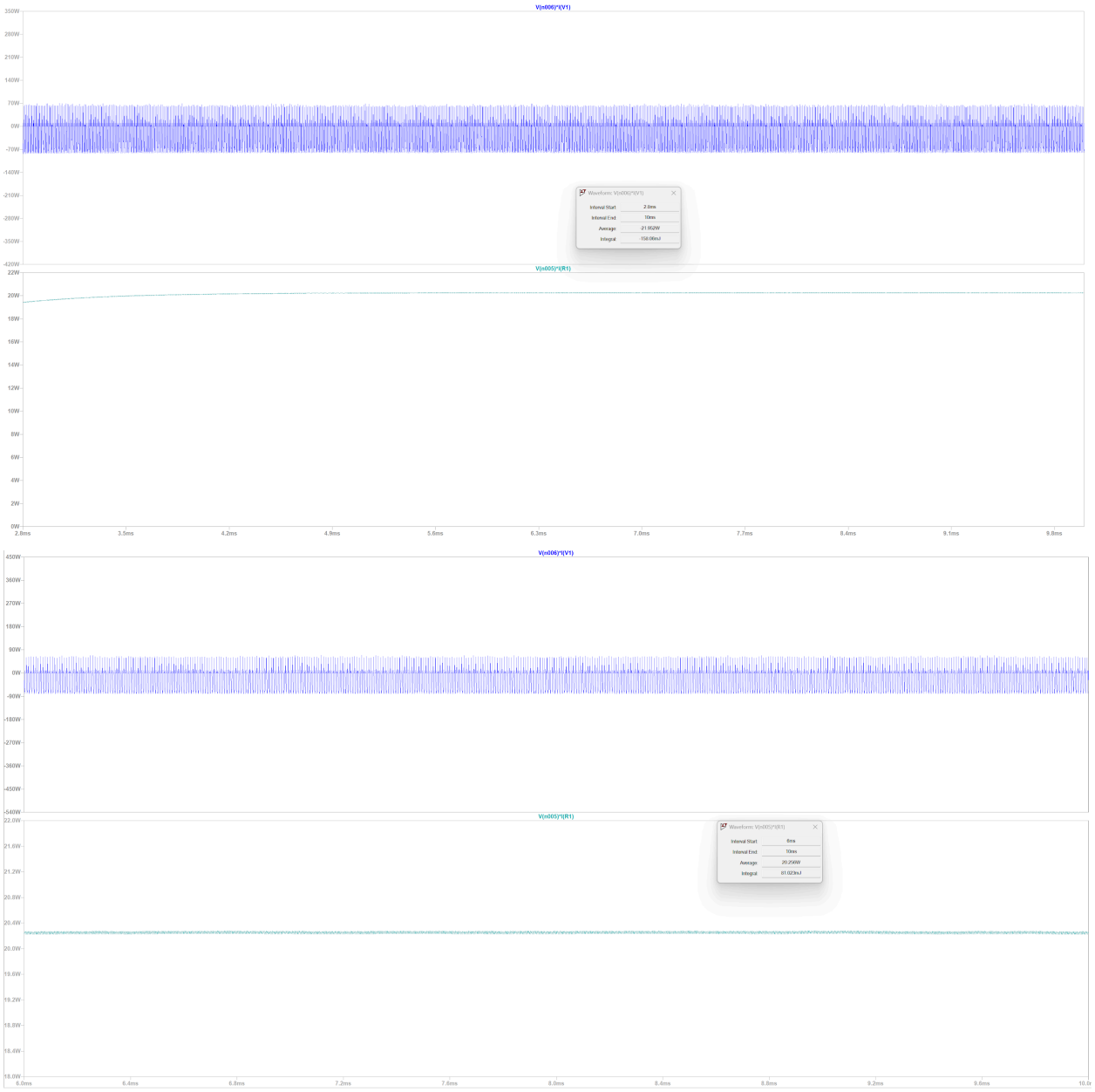

*Figures 5.5 & 5.6: Input and Output powers at 2.92Ω*



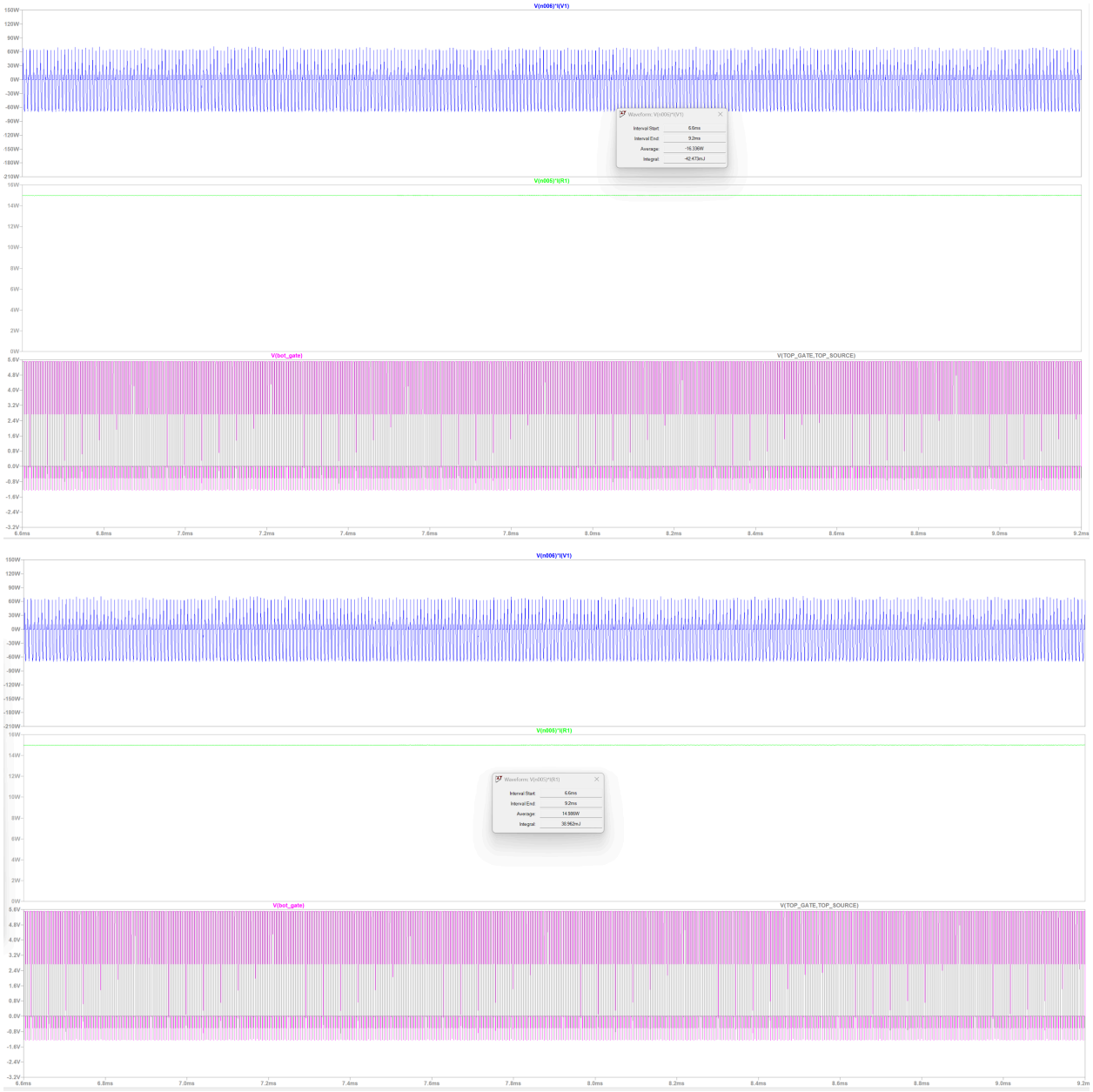

*Figures 5.7 & 5.8: Input and Output powers at 3.42Ω*




*Figures 5.9 & 5.10: Input and Output powers at 4.42Ω*



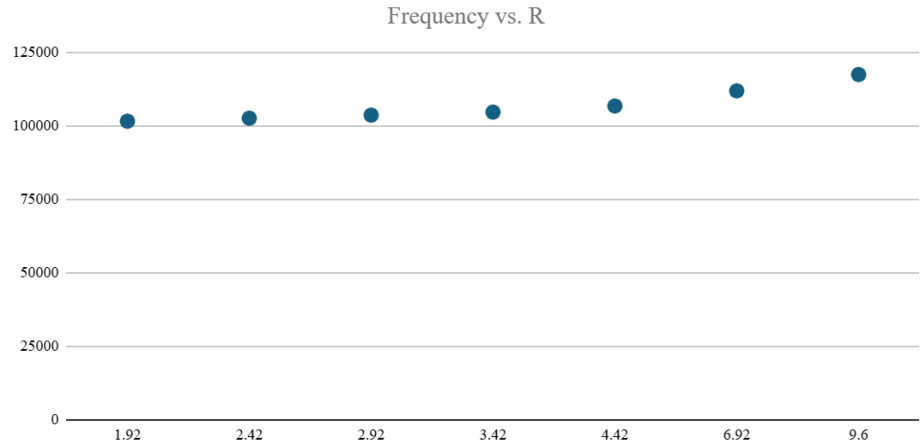

*Figures 5.11 & 5.12: Input and Output powers at 6.92Ω*



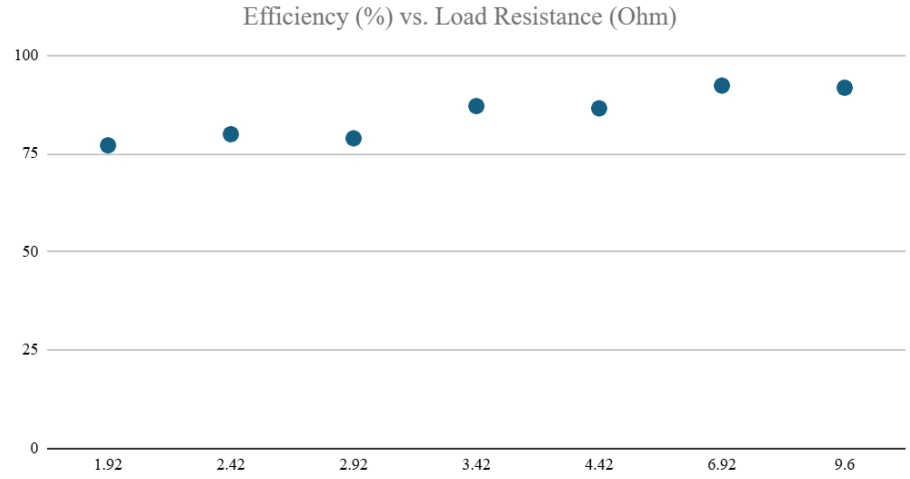

*Figures 5.13 & 5.14: Input and Output powers at 9.6Ω*

| Load Resistance (ohm) | Frequency (Hz) | Period (s) | Tdelay for bottom switch (s) | Ton (s) | Input Power (W) | Output Power (W) | Efficiency % |
|---|---|---|---|---|---|---|---|
| 1.92 | 101520 | 9.8503E-06 | 4.9251E-06 | 4.7694E-06 | 97.6450 | 75.2600 | 77.0751 |
| 2.42 | 102550 | 9.7513E-06 | 4.8757E-06 | 4.7200E-06 | 74.8660 | 59.8220 | 79.9054 |
| 2.92 | 103585 | 9.6539E-06 | 4.8270E-06 | 4.6713E-06 | 62.4160 | 49.2200 | 78.8580 |
| 3.42 | 104615 | 9.5589E-06 | 4.7794E-06 | 4.6237E-06 | 47.4400 | 41.3070 | 87.0721 |
| 4.42 | 106685 | 9.3734E-06 | 4.6867E-06 | 4.5310E-06 | 37.6070 | 32.5250 | 86.4866 |
| 6.92 | 111854 | 8.9402E-06 | 4.4701E-06 | 4.3144E-06 | 21.9520 | 20.2560 | 92.2741 |
| 9.6 | 117392 | 8.5185E-06 | 4.2592E-06 | 4.1035E-06 | 16.3360 | 14.9860 | 91.7360 |

Table: Varying loads with corresponding frequency to maintain 12V at output and converter efficiency




Plot: Frequency (Hz) vs. Load Resistance (Ohm)

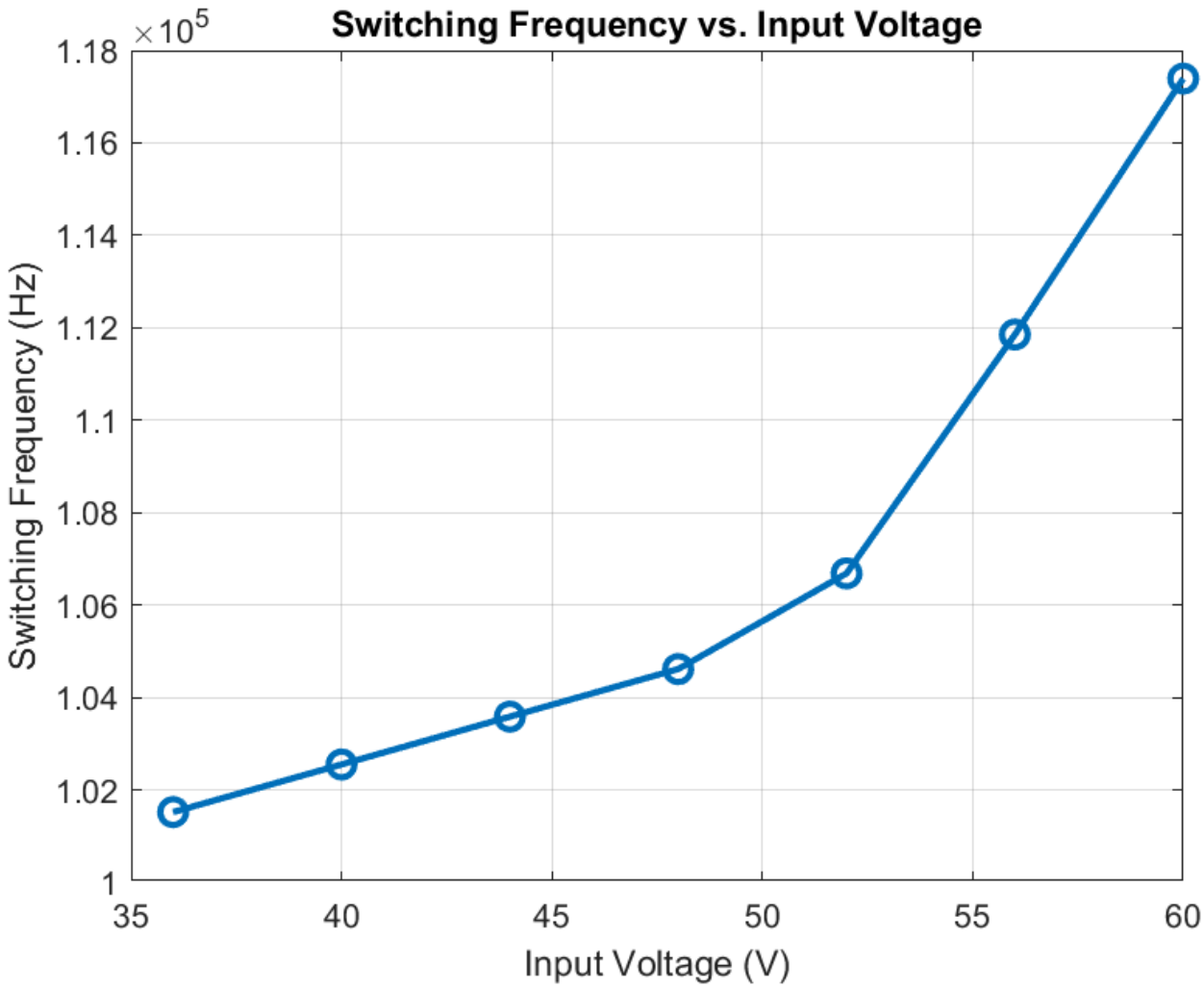


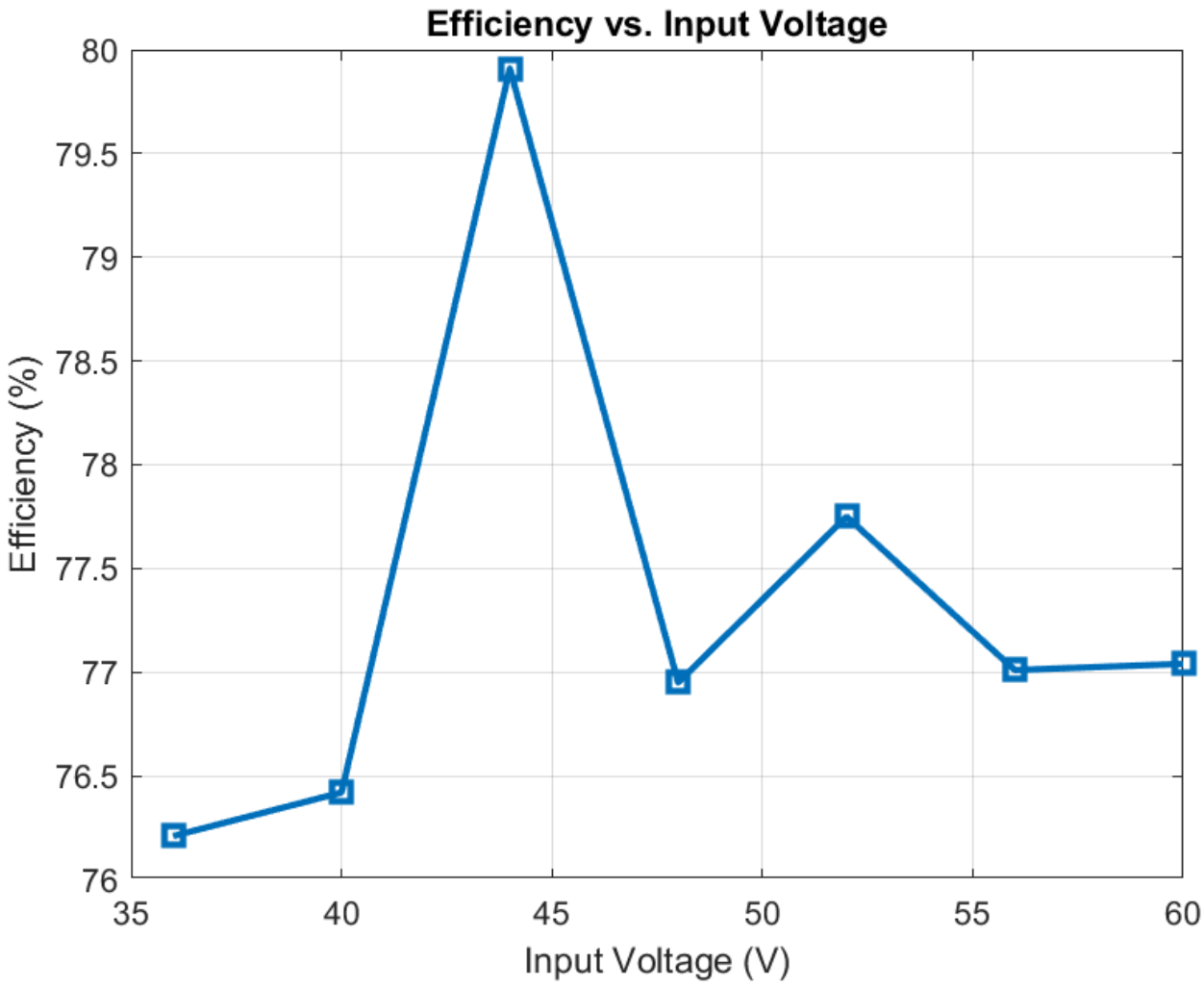

Plot: Efficiency (%) vs. Load Resistance (Ohm)

# Task 6

Use the model from Task 4 to repeat the analysis from Task 3. For RL=RL,n and selected VIN values, the frequency should be varied in the open-loop mode until the nominal output voltage is reached VO=VO,n. To achieve uniform testing, use the following input voltage values: VIN=[36 40 44 48 52 56 60] V.

In this task, various input voltages were tested out on the circuit and the efficiency was plotted with the frequency. The respective input and output power values were found, and efficiency was calculated.






The first graph indicates that the switching frequency changes with input voltage. As Vin increases, the switching frequency rises, which makes sense because the converter needs to adjust to maintain the nominal output voltage. The relationship isn't perfectly linear, meaning higher voltages require disproportionately larger frequency increases. This necessitates an effective control strategy covering a range from about **101.5 kHz to 117.4 kHz** to handle all input conditions effectively.

The second plot, showing efficiency vs. input voltage, has an interesting peak at **44V**, where efficiency reaches nearly **80%**. At other input voltages, efficiency hovers between **76% and 78%**, with a slight dip at **48V**. The spike at 44V suggests that the converter operates most efficiently at this voltage, possibly due to lower switching and conduction losses. The efficiency variations could come from increasing switching losses at higher frequencies or changes in the load conditions.

# Task 7

In this task, it is objective to determine the maximum voltage and current stress of the converter's switching devices and reactive components. Consider the four extreme operation conditions (VIN,min, RL,min), (VIN,min, RL,max), (VIN,max, RL,min), (VIN,max, RL,max). For each condition, vary the frequency and use the trial-and error approach to set the output voltage to its nominal value. Considering all four cases: Determine the extreme values of the parameters listed in the table below and include the table in your report

Does the converter design satisfy the necessary conditions for ZVS operation? Using the switch specification, investigate if sufficient ZVS conditions are reached for each case (for the analysis, review the discussion in Topic 1 and adopt VB=0.7 V). For which of the cases is the half-bridge operation closest to hard-switching mode? Illustrate the ZVS operation for the most critical case by plotting vinv and iinv, the two gate signals, and the individual currents of the top and bottom switch during the switching instant. Explain how the signal shape indicates the ZVS operation of the half bridge. Include the calculations, plots, and discussion in your report.

| Signal | Unit | Value |
|---|---|---|
| Maximum of the inverter current Iinv | A | 11.434 |
| Maximum of the resonant capacitors' voltage Vr | V | 436.122 |
| Maximum of the primary voltage Vp | V | 19.35 |
| Maximum of the rectified current IR | A | 20.44 |
| Maximum output voltage ripple vO,ripp | % | 75.44 |
| Minimum switching frequency fmin | Hz | 102.42k |
| Maximum switching frequency fmax | Hz | 118.13k |
| The minimum phase of the resonant network impedance Zinv | deg. | 0.2532 |
| The maximum phase of the resonant network impedance Zinv | deg. | 11.148 |

From the observations of this experiment, it can be determined that the converter satisfies the requirements for Zero-Voltage Switching (ZVS) in the cases of Vin max, RL max and Vin min, RL max. In these scenarios, the phase angle is at its minimum, indicating operation at or very close to the resonant frequency. This condition ensures that the current lags the voltage, thereby facilitating ZVS operation, where the switches turn on at near-zero voltage, reducing switching losses and improving overall efficiency. The experimental data further confirms that under these conditions, the converter maintains its intended soft-switching behavior, minimizing stress on switching components.

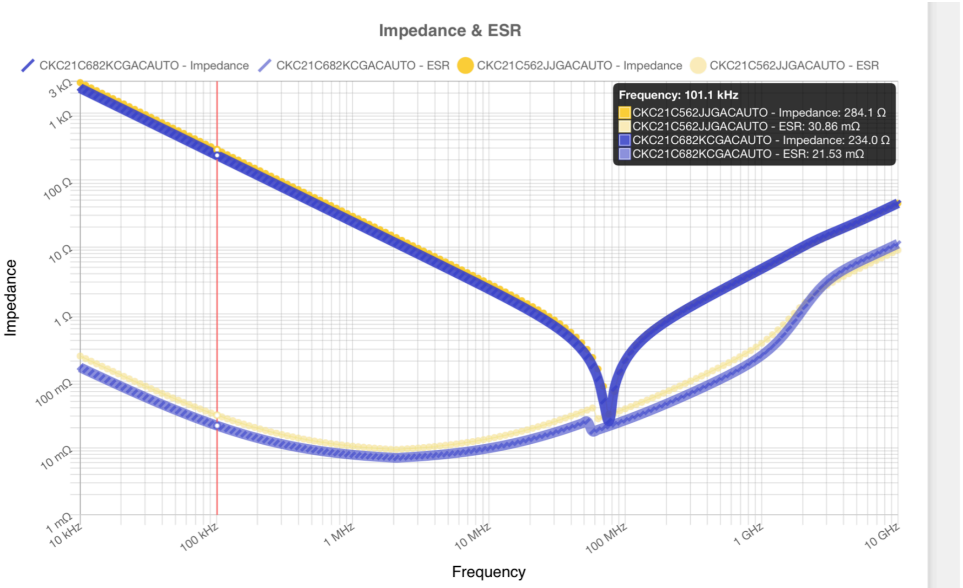
In contrast, for the case of Vin max, RL min, the phase angle is at its maximum, making the operation closest to hard switching. A potential reason for this behavior is the mismatch between the input voltage and the load resistance. The interplay between these two parameters significantly impacts the resonance condition of the converter. When the load resistance is at its minimum while the input voltage remains high, the converter experiences a deviation from its ideal resonant operation, leading to increased voltage stress across the switches during turn-on. This condition shifts the operation away from ZVS and towards hard switching, resulting in higher switching losses. Thus, achieving optimal soft-switching performance requires careful balancing of the input voltage and load conditions to maintain resonance and minimize phase deviations.

# Appendix A

Task 1,2,3: The MATLAB code for TASK 1, 2, 3 is added in different zipped files which are submitted with the report.

Task 4: Equivalent series resistance reference for resonant capacitors

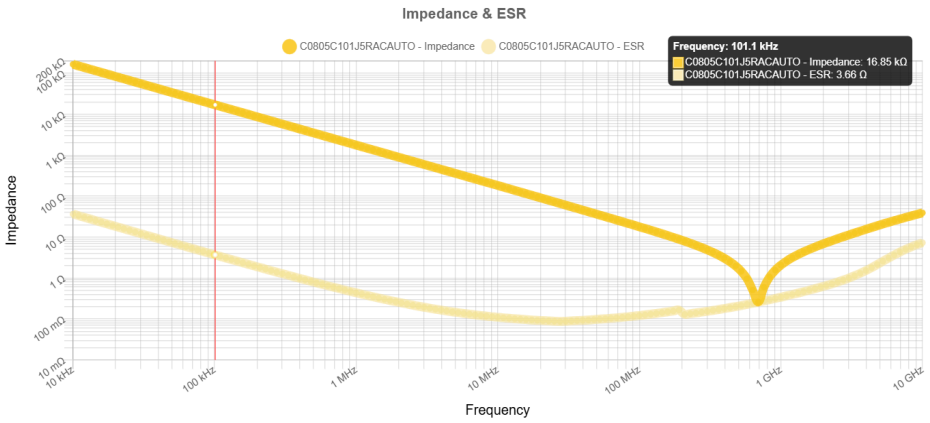
| | |
|---|---|
| Resonant Capacitance $C_r$ | 4 x 5.6 nF<br>‖<br>3 x 6.8nF<br>Total:42.8 nF |




| Info | Legend | Tolerance | Part Number | Cap. | $V_{DC}$* | Dielectric | Qty. | Bias (V) | Amb. (°C) | Add | Remove |
|---|---|---|---|---|---|---|---|---|---|---|---|
| › | ● | 10% | Ckc21C562jjgacauto | 5.6 nF | 1700 | C0G | 1 | 0 | 25 | + | × |
| › | / | 10% | **CKC21C682KCGACAUTO** | **6.8 nF** | **500** | **C0G** | 1 | 0 | 25 | + | × |

Task 4: Equivalent series resistance reference for filter capacitors

| | |
|---|---|
| Filter Capacitors $C_f$ and $C_c$ | 0.1 nF<br>10 nF<br>1 uF<br>5 x 33uF<br>165 uF |




| Info | Legend | Tolerance | Part Number | Cap. | $V_{DC}$* | Dielectric | Qty. | Bias (V) | Amb. (°C) | Add | Remove |
|---|---|---|---|---|---|---|---|---|---|---|---|
| › | | 10% | C0805C101J5RACAUTO | 100 pF | 50 | X7R | 1 | 0 | 25 | + | × |

Product Index › Capacitors › Aluminum - Polymer Capacitors › KEMET A767KN336M1VLAE031

**A767KN336M1VLAE031**



*Image shown is a representation only. Exact specifications should be obtained from the product data sheet.*

| | |
|---|---|
| DigiKey Part Number | 399-A767KN336M1VLAE031TR-ND - Tape & Reel (TR)<br>399-A767KN336M1VLAE031CT-ND - Cut Tape (CT)<br>399-A767KN336M1VLAE031DKR-ND - Digi-Reel® |
| Manufacturer | KEMET |
| Manufacturer Product Number | A767KN336M1VLAE031 |
| Description | CAP ALUM POLY 33UF 20% 35V SMD |
| Manufacturer Standard Lead Time | 26 Weeks |
| Customer Reference | |
| Detailed Description | 33 µF 35 V Aluminum - Polymer Capacitors Radial, Can - SMD 31mOhm 2000 Hrs @ 105°C |
| Datasheet | Datasheet |
| EDA/CAD Models | A767KN336M1VLAE031 Models |

**Product Attributes**

| TYPE | DESCRIPTION | SELECT ALL |
|---|---|---|
| Category | Capacitors<br>Aluminum - Polymer Capacitors | |
| Manufacturer | KEMET | |
| Series | A767 | |
| Packaging | Tape & Reel (TR)<br>Cut Tape (CT)<br>Digi-Reel® | |
| Part Status | Active | |
| Type | Polymer | |
| Capacitance | 33 µF | |
| Tolerance | ±20% | |
| Voltage - Rated | 35 V | |
| ESR (Equivalent Series Resistance) | 31mOhm | |
| Lifetime @ Temp. | 2000 Hrs @ 105°C | |
| Operating Temperature | -55°C ~ 105°C | |
| Ratings | - | |
| Applications | General Purpose | |
| Features | - | |

KYOCERA AVX Part No.

# KGM21NR71H103JM

Hist.PN: 08055C103JAT4A

CERAMIC DIVISION

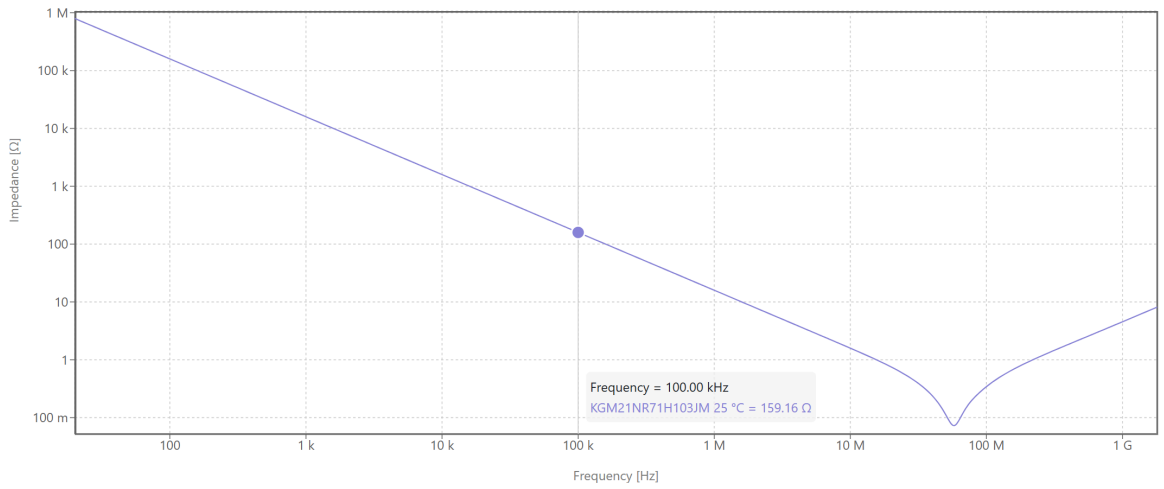
Category - General

DOWNLOAD DATASHEET - KGM21NR71H103JM | DOWNLOAD | CHECK INVENTORY

X7R - General

| Case Size [EIA] | Capacitance [F] | Tolerance | Voltage [V] | Dielectric | Termination | Product Type | Technology |
|---|---|---|---|---|---|---|---|
| 0805 | 10 n | ± 5 % | 50 | X7R | Sn | General | Mlcc |

IMPEDANCE | ESR | TEMPERATURE | DC BIAS | RIPPLE CURRENT | EQ CIRCUIT DIAGRAM | CASE DIMENSIONS




*Typical values only*

# Appendix B
# Task 6 MATLAB code:

```
%%
clc;
clear all;
close all;
% Data from the spreadsheet
Vin = [36 40 44 48 52 56 60]; % Input Voltage (V)
Frequency = [101520 102550 103585 104615 106685 111854 117392]; % Switching Frequency (Hz)
Efficiency = [76.21 76.42 79.91 76.95 77.75 77.01 77.04]; % Efficiency (%)
% Plot Switching Frequency vs. Input Voltage
figure;
plot(Vin, Frequency, 'o-', 'LineWidth', 2, 'MarkerSize', 8);
xlabel('Input Voltage (V)');
ylabel('Switching Frequency (Hz)');
title('Switching Frequency vs. Input Voltage');
grid on;
% Plot Efficiency vs. Input Voltage
figure;
plot(Vin, Efficiency, 's-', 'LineWidth', 2, 'MarkerSize', 8);
xlabel('Input Voltage (V)');
ylabel('Efficiency (%)');
title('Efficiency vs. Input Voltage');
grid on;
```